BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2024

BIOLOGY

Past Paper 2024 • Paper Code: PR–XII (A-I-A) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** PR–XII (A-I-A) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Blastocyst is attached to the uterus by:

(A) Inner cell mass
(B) Epibalst
(C) Hypoblast
(D) Trophoblast

Q2. In 4 O'clock plant R2R2 represent ____ flower.

(A) Yellow
(B) Red
(C) White
(D) Pink

Q3. Human blood group ABO are controlled by____number of alleles.

(A) 1
(B) 2
(C) 3
(D) 15

Q4. According to conservative model the parental DNA remain____ After replication.

(A) Intact
(B) Dispersed
(C) Unwind
(D) Hybrid

Q5. Genetic codes, those encodes specific amino acids are termed as:

(A) Anticodes
(B) Stop codons
(C) Start codons
(D) Sense codons

Q6. Organisms evolved through the inheritance of acquired characteristics according to:

(A) Lamark
(B) Lyell
(C) Cuvier
(D) Louis Agassiz

Q7. The rate at which the new individuals are added to a population in a unit time is:

(A) Mortality
(B) Natality
(C) Dispersion
(D) Succession

Q8. Genome mean____ gene of an organism.

(A) Single
(B) Few
(C) No
(D) All

Q9. Chimpanzee and Gorilla can be categorized as____animals.

(A) Draught
(B) Tamed
(C) Wild
(D) Domesticated

Q10. Respirator surfaces must be:

(A) Thick
(B) Hard
(C) Thin
(D) Impermeable

Q11. Long term exposure to industrial pollutants or dust may cause:

(A) Emphysema
(B) Empyema
(C) Hyponatremia
(D) Tinnitus

Q12. The capillaries of vasa recta run anti-parallel to the:

(A) Glumerulus
(B) Bowman's capsule
(C) Loop of Henle
(D) Collecting duct

Q13. Sphenoid and Ethmoid bones are the part of human:

(A) Neck
(B) Skull
(C) Trunk
(D) Limbs

Q14. Sciatica refer to a pain in the:

(A) Spine
(B) Joint
(C) Bone
(D) Leg

Q15. Nerve fiber is different from muscle fiber due to the presence of:

(A) Sarcomere
(B) Sarcolemma
(C) Dendrites
(D) Myofibrils

Q16. Testosterone is secreted from:

(A) Gonades
(B) Ovaries
(C) Testes
(D) Kidneys

Q17. In animals hostile or friendly interactions are examples of ____ behaviors.

(A) Social
(B) Innate
(C) Habituation
(D) Imprinting

Q18. In females cyclic menstruation is an indicator of:

(A) Infertility
(B) Menopause
(C) Normal reproduction
(D) Endometriosis

SECTION-B

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 40

Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FOUR marks.

1. State about the volumes and capacities of lungs.
2. Compare the structure and function of nephron and dialyzer?
3. Enlist the name of joints according to their structural classification.
4. Write briefly about drug tolerance.
5. Define hormone and name the hormones secreted from anterior lobe of pituitary gland?
6. Innate behavior ensure the survival of individual in a population, how?
7. Briefly discuss about Epididymis and Fallopian tubes.
8. Mention about the system level changes in aging?
9. Illustrate about Erythrobustosis foetalis.
10. Write any four differences between Transcription and Translation.
11. Briefly elaborate the statement "Natural Selection or Survival of the fittest".
12. Briefly discuss about Global Warming.
13. Write the applications of PCR.

SECTION-C

Time Allowed: 2 Hours 40 Minutes | **Maximum Marks:** 27

> Note: Note: Attempt any THREE questions. All questions carries equal marks.

Q. Attempt any THREE questions. All questions carries equal marks.

Q3.

(a) Explain the term Acclimatization and Selective Breeding.

(b) Mention any four aspects related to the safety in Biotechnology.

Q4.

(a) Discuss about human lungs.

(b) Discuss about lithortripsy.

Q5.

(a) Write the causes, kinds and symptoms of stroke.

(b) Briefly explain negative feedback mechanism.

Q6. Explain gene and chromosomal mutation in detail.